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MEDGENOME

DNA TEST REPORT

Full Name / Ref No	**Shaurya Nayak (L18662-726)**	Order ID/Sample ID	**1749426/9957238**
Date of Birth / Age	**4 Years and 3 Months**	Gender	**Male**
Parental Sample ID	**NA**	Sample Type	**Peripheral Blood in EDTA (Purple Top)**
Referring Clinician	**Dr. Biren Bhatt Toprani Advanced Lab Systems Vadodara**	Date and time of Sample Collection	**02-05-2026; 12.00 AM**
		Date and time of Sample Receipt	**03-05-2026; 12.09 PM**
		Date and time of Order	**04-05-2026; 11.18 AM**
		Date and time of Report	**19-05-2026; 12.00 AM**
Test Requested	**DUCHENNE MUSCULAR DYSTROPHY (DMD) DELETION/DUPLICATION ANALYSIS [MGM136]**		

CLINICAL DIAGNOSIS / SYMPTOMS / HISTORY

The patient, presented with clinical indications of calf hypertrophy, difficulty in getting up from sitting position, leg pain and an elevated CPK level of 22,051U/L. The patient is thus being evaluated for large pathogenic deletions/duplications in *DMD* gene.

RESULTS

PATHOGENIC VARIATION ASSOCIATED WITH THE SUSPECTED PHENOTYPE WAS IDENTIFIED

Sl. No.	Deletions /Duplications	No. of exons deleted/duplicated ‡	MLPA probe ratio (Dosage quotient) #	Disease (OMIM)	Inheritance	Classification
1	Hemizygous deletion	1 Exon (50)*	0.00	Becker muscular dystrophy / Duchenne muscular dystrophy	X-linked recessive	Pathogenic

ADDITIONAL FINDINGS: NO VARIANT(S) OF UNCERTAIN SIGNIFICANCE (VUS) IDENTIFIED

CLINICAL CORRELATION AND VARIANT INTERPRETATION

- Hemizygous deletion was detected within the detection limits of MLPA, in *DMD* gene in this subject.
- This is an OUT-OF-FRAME deletion, as confirmed using the reading frame checker (2). According to the reading frame rule this variation in DMD is associated with a DMD phenotype. However, exceptions to the reading frame rule are seen in 10% of the cases (3,4).
- The deletion/duplications in *DMD* gene have been previously identified in patients with Duchenne muscular dystrophy / Becker muscular dystrophy.
- Based on the above evidence, the variation detected in this subject is classified as pathogenic and must be carefully correlated with the clinical symptoms observed.

RECOMMENDATIONS

Genetic counselling is advised.
*Point variations / small INDELS in probe binding region may result in single exon deletion profiles in MLPA. Hence, sequencing of exon 50 of the DMD gene is recommended to rule out point variations. Screening of the mother is recommended to determine the carrier status.

BACKGROUND

Duchenne muscular dystrophy (DMD) is a recessive, X-linked disorder caused by mutations in the dystrophin (DMD) gene located on the human X chromosome (Xp21.2). The gene encodes a protein dystrophin that is a part of the dystroglycan complex (DGC), which provides structural stability to the muscle tissue. The disorder characterized by progressive muscle weakness and wasting (atrophy) occurs at a frequency of about 1 in 3,500 almost exclusively in new-born males and results in muscle degeneration over time. More than 1,000 mutations in the *DMD* gene have been identified in people with the Duchenne and Becker forms of Muscular Dystrophy. Most of the mutations that cause DMD delete or duplicate part of the *DMD* gene will prevent any functional dystrophin protein from being produced

TEST METHODOLOGY

Copy number changes in targeted regions of the *DMD* gene were identified by hybridizing with MLPA (Multiplex Ligation dependent Probe Amplification) probes. Each MLPA probe consists of two hemi-probes that bind to adjacent sites on the target sequence. Upon ligation and subsequent PCR amplification, each distinct MLPA probe (specific to distinct target regions) generates an amplicon with a unique length which are separated and quantified by capillary electrophoresis. Deletions of a probe's recognition sequence on the X-chromosome will lead to a complete absence of the corresponding probe amplification product in males. Heterozygous deletions within target sequences will prevent efficient probe binding and give a 35-50% reduced relative peak area of the amplification product specific to that probe set. Copy number differences of various exons between test and control DNA samples can be detected by analyzing the MLPA peak patterns.

*** Genetic test results are reported based on the recommendations of American College of Medical Genetics (Richards CS et al., Genet Med, 2015), as described below:**

Variant	A change in a gene. This could be disease causing (pathogenic) or not disease causing (benign).
Pathogenic	A disease causing variation in a gene which can explain the patients' symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed.
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Benign	A variant which is known not to be responsible for disease has been detected. Generally no further action is warranted on such variants when detected.
Likely Benign	A variant is not expected to have a major effect on disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion.
Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non-disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

#The exon numbering is based on the DMD mRNA reference sequence NM_004006.2 nomenclature in the NCBI GenBank database, which represents transcript variant Dp427m. It encodes the main dystrophin protein found in the muscle.

#MLPA ratios (dosage quotient) of below 0.7 or above 1.3 are indicative of a deletion (copy number change from two to one) or duplication (copy number change from two to three), respectively. A dosage quotient of 0.0 indicates a homozygous/hemizygous deletion, 0.35 to 0.65 indicates heterozygous deletion, 1.35 to 1.55 indicates heterozygous duplication and 1.7 to 2.2 indicates homozygous/hemizygous duplication. A MLPA ratio (dosage quotient) between 0.80 to 1.20 indicates a normal copy number status.

DISCLAIMER

- The MLPA test will not detect the point mutations in the *DMD* gene, which are the second most common cause of genetic defects in the *DMD* gene. It is therefore recommended to use MLPA in combination with sequence analysis.
- A point mutation or polymorphism in the sequence detected by a probe, which results in reduced probe binding efficiency, can also cause a reduction in relative peak area. Therefore, single exon deletions detected by MLPA should always be confirmed by other methods like multiplex PCR or sequencing.
- MLPA cannot detect any changes that lie outside the target sequence of the probes and will not detect most inversions or translocations. Even when MLPA did not detect any aberrations, the possibility remains that biological changes in that gene or chromosomal region do exist but remain undetected.

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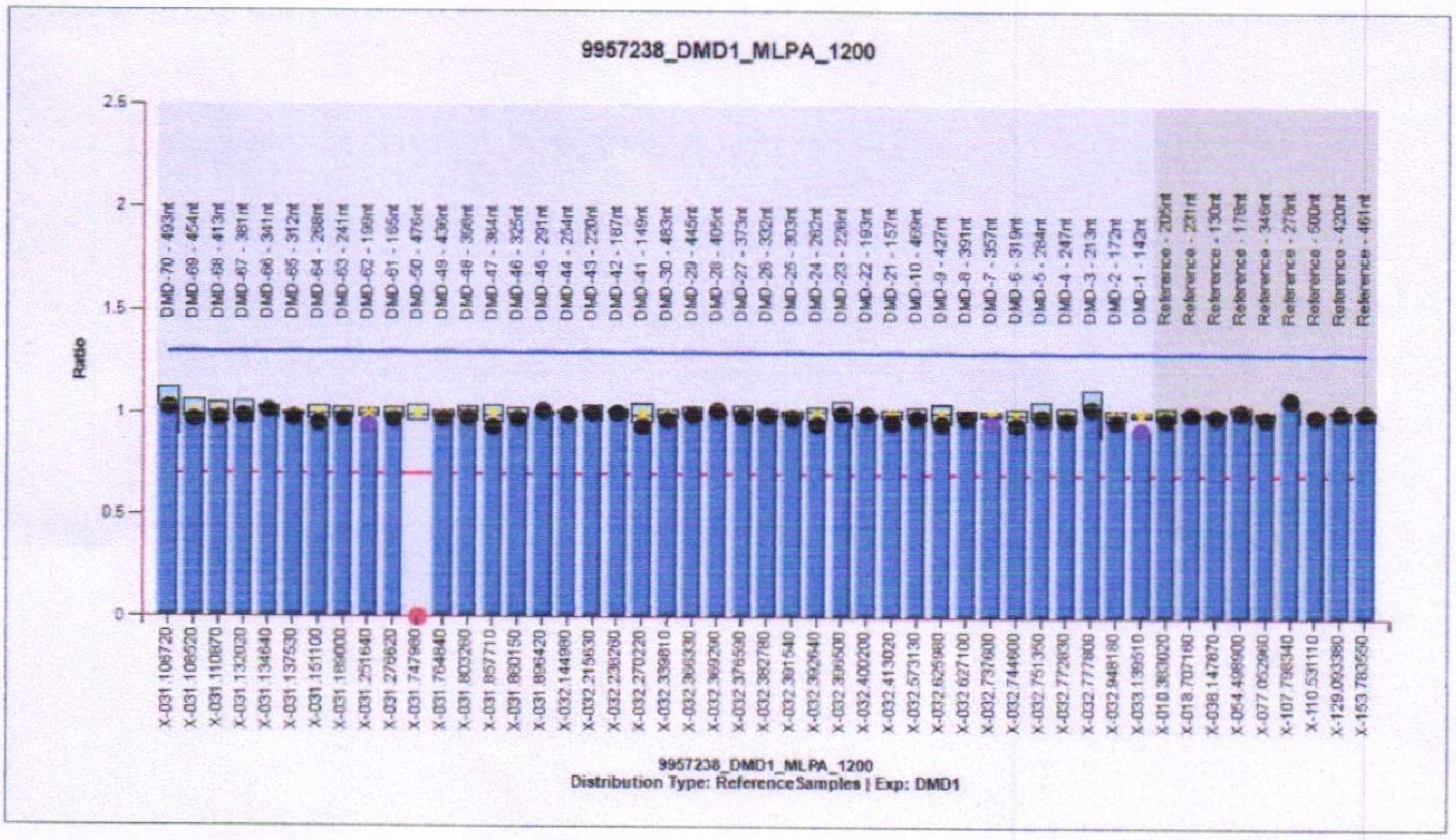
REFERENCES

1. Wang X et al. Similarity of *DMD* gene deletion and duplication in the Chinese patients compared to global populations. Behavioural and Brain Functions 2008, 4:20 .doi:10.1186/1744-9081-4-20.
2. www.dmd.nl
3. Jean K. Mah et al, A population-based study of dystrophin mutations in Canada. Can J Neurol Sci. 2011 May; 38(3):465-74.
4. Kevin M. Flanigan et. al. Mutational Spectrum of DMD Mutations in Dystrophinopathy Patients: Application of Modern Diagnostic Techniques to a Large Cohort. Hum Mutat. 2009 December; 30(12): 1657–1666. doi:10.1002/humu.21114.
5. S.J. White et al, Duplications in the *DMD* Gene. HUMAN MUTATION 27(9), 938-945, 2006

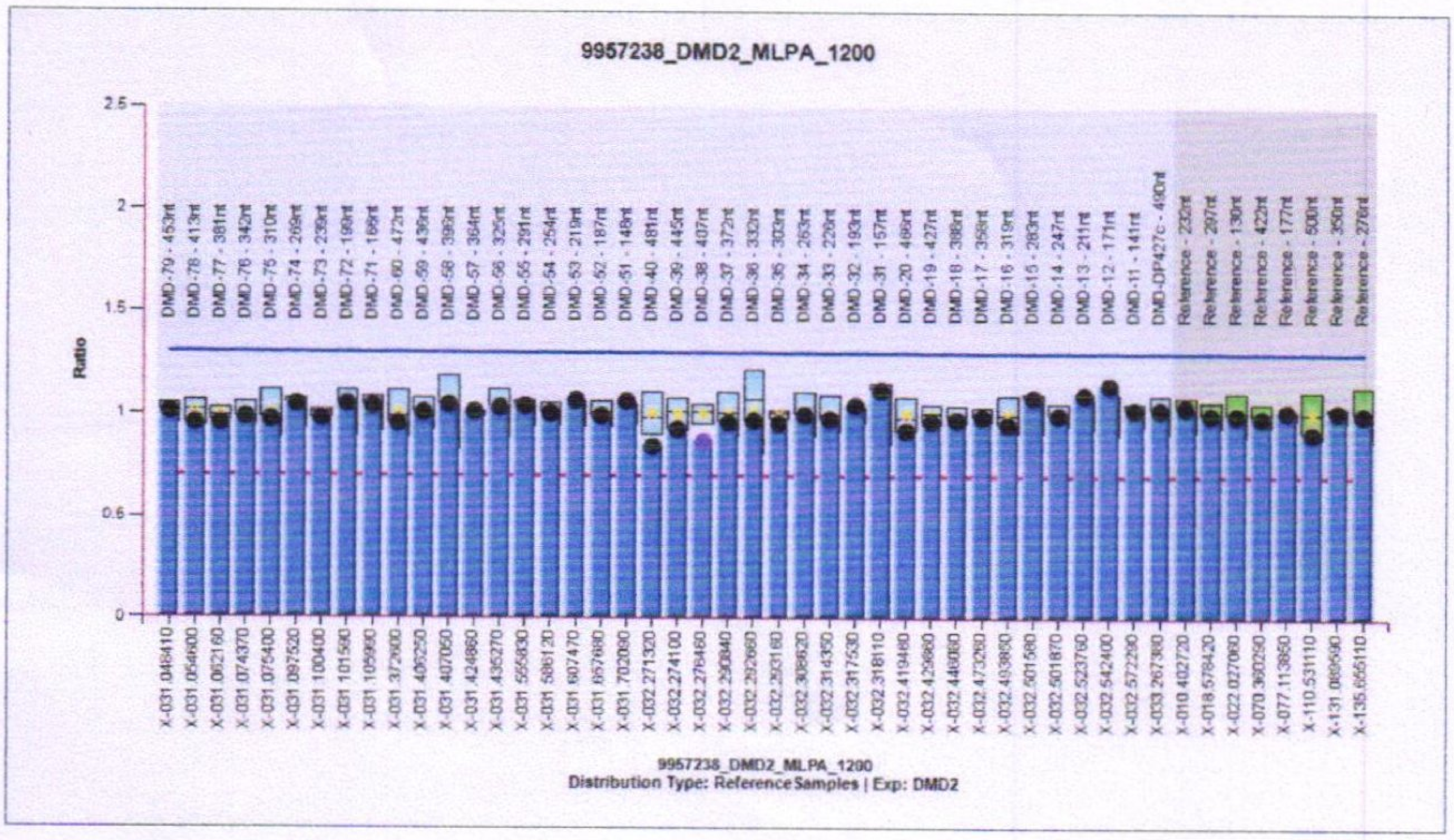
APPENDIX-1

DMD-MLPA Result Figures

MLPA Ratio Chart: DMD probemix 1



MLPA Ratio Chart: DMD probemix 2



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